

Clean Code Development Cheat Sheet

1. Use Meaningful Names

Bad

```
const d = getData();
```

Good

```
const matchStatistics = getMatchStatistics();
```

2. Prefer Composition Over Inheritance

Bad

```
class PremiumPredictionEngine  
  extends AdvancedPredictionEngine
```

Good

```
class PredictionService {  
  constructor(  
    private predictionModel: PredictionModel  
  ) {}  
}
```

3. Single Responsibility Principle

Bad

```
async function generatePrediction(matchId: string) {  
  const data = await fetchMatchData(matchId);  
  const model = await loadModel();  
  const prediction = model.predict(data);  
  
  await sendNotifications(prediction);  
}
```

Good

```
async function generatePrediction(matchId: string) {  
  const matchData = await matchService.getMatchData(matchId);  
  const prediction = await predictionService.calculatePrediction(matchData);  
  const notifications = await notificationService.sendNotifications(prediction);  
}
```

4. Reuse logic i.e. no code duplication

Bad

```
calculateHomeWinRate();
calculateAwayWinRate();
```

Good

```
calculateWinRate(team);
```

5. Define numbers

Bad

```
if (confidence > 0.75)
```

Good

```
const MIN_CONFIDENCE_THRESHOLD = 0.75;
```

6. Return early

Bad

```
function predict(match: Match) {
  if (match) {
    if (match.players) {
      return model.predict(match);
    }
  }
}
```

Good

```
function predict(match: Match) {
  if (!match.players) {
    return null;
  }

  return model.predict(match);
}
```

7. Use Explicit Types

Bad

```
function predict(data: any) {}
```

Good

```
function predict(data: MatchFeatures): PredictionResult {}
```

8. Use Enums (instead of Strings)

Bad

```
if (match.status === "finished") {  
    generateReport();  
}
```

Good

```
enum MatchStatus {  
    Scheduled,  
    Live,  
    Finished  
}  
  
if (match.status === MatchStatus.Finished) {  
    generateReport();  
}
```

9. Handle Errors Properly

Bad

```
catch(error) {}
```

Good

```
catch(error) {  
    logger.error(error);  
}
```

10. Use Constructors

Bad

```
const repository = new PredictionRepository();
```

Good

```
constructor(  
    private repository: PredictionRepository  
) {}
```